BOARD QUESTION PAPER:
MARCH 2016 Physics

Time: 3 Hours Total Marks: 70

Note:

(i) All questions are compulsory.

(ii) Neat diagrams must be drawn wherever necessary.

(iii) Figures to the right indicate full marks.

(iv) Use of only logarithmic tables is allowed.

(v) All symbols have their usual meaning unless otherwise stated.

(vi) Answer to both sections must be written in the same answer book.

(vii) Answer to every question must be written on a new page.

SECTION -I

Question 1. Attempt any SIX **[12]**

(i) *In U.C.M (uniform* circular motion), prove that the relation $\vec{v} = \vec{\omega} \times \vec{r}$ **, where symbol have their usual meaning.**

(ii) Derive an expression for critical velocity of satellite revolving around the sun the earth in circular orbit.

(iii) Obtain an expression for total kinetic energy of a rolling body in for $\frac{1}{2}MV^2\left[1+\frac{K^2}{R^2}\right]$.

(iv) Define 'emissive power 'and 'coefficient of emission of a body'.

(v) A coin kept at a distance of 5cm from the center of turntable of radius 1.5m just begains to slip when the turntable rotates at speed of 90r.p.m. Calculate the coefficient of static friction between the coins the turntable. $\left[g = 9.8m/s^2\right]$

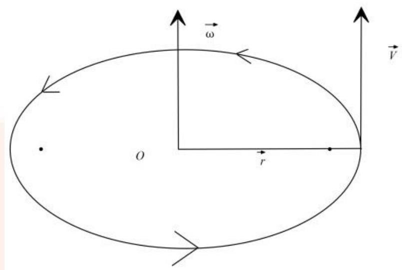
(vi) The fundamental frequency of an air column in a pipe closed at one end is in unison with the third overtone of an open pipe. Calculate the ratio of their lengths of their air columns.

(vii) A particle performing linear SHM has a period of 6.28 seconds and a path length of 2cm. What is the velocity when its displacement is 6cm from mean position?

(viii) The energy of the free surface of a liquid drop is 5π **times the surface tension of the liquid. Find the diameter of the drop in C.G.S. system.**

Solution 1:

(i) We have to prove $\vec{v}=\vec{\omega}\times\vec{r}$,



For, U.C.M we know the relation,

$$\delta\vec{S}=\delta\vec{\theta}\times\vec{r}$$

Divide by δt in both sides then we get,

$$\frac{\delta\vec{S}}{\delta t}=\frac{\delta\vec{\theta}}{\delta t}\times\vec{r}$$

Now, by taking limit $\delta t\xrightarrow{\lim}0$ on both sides,

$$\lim_{\delta t\to 0}\frac{\delta\vec{S}}{\delta t}=\lim_{\delta t\to 0}\frac{\delta\vec{\theta}}{\delta t}\times\vec{r}$$

$$\frac{\delta\vec{S}}{\delta t}=\frac{\delta\vec{\theta}}{\delta t}\times\vec{r}..............(1)$$

Since,

$$\frac{d\vec{S}}{dt}=\vec{V}.....(2)$$

$$\frac{d\vec{\theta}}{dt}=\vec{\omega}.....(3)$$

NOw from eq$(1),(2)$ and (3) we get

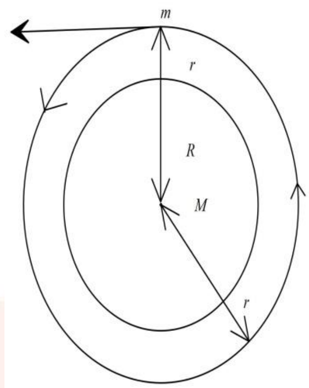
$$\vec{v}=\vec{\omega}\times\vec{r}$$

Hence, $\vec{v}=\vec{\omega}\times\vec{r}$ proved

(ii) **Critical velocity: -** It is defined by the minimum horizontal velocity of projection that must be provided satellite at a certain height. Such that it travels in the circular orbit around the earth.

Let us consider a satellite of mass 'm' revolving around the earth at a height 'h' from the surface.

Suppose 'M' be the mass of the Earth and 'R' be the radius.



Here, the satellite is moving in circular orbit with critical velocity 'Vc' and radiusof orbit is $r = R + h$.

The centripetal force for the circular motion is given by the circular motion.

Therefore,

Centripetal force is equal to gravitational force

$$\frac{mV_c^{\ 2}}{r} = \frac{GMm}{r^2}$$

$$V_c^2 = \frac{GM}{r}$$

$$V_c = \sqrt{\frac{GM}{r}}$$

$$V_c = \sqrt{\frac{GM}{R+h}}$$

Therefore, this is the expression of critical velocity $V_c = \sqrt{\frac{GM}{R+h}}$.

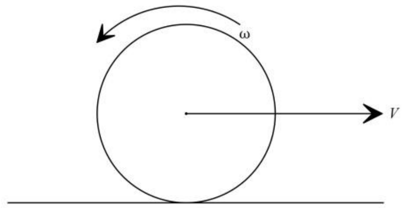
Here

$R + h \approx R$

Hence,

$$V_c = \sqrt{\frac{GM}{R}}$$

(iii)



Here, a body is rolling on a surface the motion can be treated as an combination of both translator motion of center of mass.

Therefore, the total kinetic energy of motion is $E = E_T + R_R ....(I)$

Where, E_T is the translational kinetic energy and E_R is the rotational energy

Let M be the mass and R be the radius V be the translational speed, ω be the angular speed.

I be the moment of inertia and K be the radius of gyration

So,

$$E_T = \frac{1}{2}mv^2.....(2)$$

$$E_R = \frac{1}{2}I.\omega^2...(3)$$

Therefore, from eq(1) (2) and (3) we get,

$$E = E_T + E_R$$
$$= \frac{1}{2}mv^2 + \frac{1}{2}I.\omega^2$$
$$= \frac{1}{2}mv^2 + \frac{1}{2}Mk^2\omega^2 \quad \left(I = mk^2\right)$$
$$= \frac{1}{2}mv^2 + \frac{1}{2}Mk^2\frac{v^2}{R^2} \quad \left(\omega = \frac{v}{R}\right)$$
$$\therefore E = \frac{1}{2}Mv^2\left(1 + \frac{K^2}{R^2}\right)$$

(iv) Emission power is the power at the given temperature is the quantity of radiant energy emitted by the body per unit time per unit surface area of the body at that temperature.

If Q is the amount of radiant energy emitted, A is the surface area of the body and t is the time for which body radiates energy. $E = \frac{Q}{At}$

Now, the coefficient of emission of a body is the ratio of the emissive power, Coefficient of emission is $e = \frac{E}{E_b}$.

(v) We have,

$$r = 5cm = 0.05m,$$
$$n = 90r.p.m = 15Hz$$

Limiting force is centrifugal force,

$$\mu_s mg = mr\omega^2$$
$$\therefore \mu_s = \frac{r\omega^2}{mg}$$
$$= \frac{r(2\pi n)^2}{g}$$
$$\therefore \mu_s = \frac{4\pi^2 n^2}{g}$$
$$= \frac{4\times 0.05\times (3.14)^2 \times (1.5)^2}{9.8}$$
$$= 0.4527$$

(vi) We have,

$$n_3 = n_0$$

Where, n_3 is the frequency of the third overtone of the open pipe and n_0 is the fundamental frequency of the closed pipe.

So, the fundamental frequency of closed pipe

$$n_0 = \frac{v}{4L_0}$$

$$\frac{v}{4L_0} = 4\left(\frac{v}{2L_3}\right)$$
$$\therefore \frac{L_0}{L_3} = \frac{1}{8}$$
$$\therefore L_0 : L_3 = 1:8$$

Hence, the ratio of the lengths of air column in the closed and open pipes is $1:8$.

(vii) We have,

$T = 6.28s$

Path length(l) is 20cm.

$2A = 20$

Amplitude $(A) = 10cm$

So,

The angular speed of the body performing S.H.M is given by

$$\omega = \frac{2\pi}{T}$$
$$= \frac{2\pi}{6.28}$$
$$\omega = 1rad \quad (\pi = 3.142).....(1)$$

Therefore, the velocity of the body at displacement 'x' from the mean position is

$$v = \omega\sqrt{A^2 - x^2}$$
$$v = 1\sqrt{(10)^2 - (6)^2} \qquad \text{x=6cm}$$
$$\text{v=}\sqrt{100 - 36}$$
$$= \sqrt{64}$$
$$v = \pm 8cm/s$$

Hence, the required velocity of the body is $\pm 8cm/s$.

(viii) We have the surface drop of a liquid is 5π times

So, the relation between surface energy and surface tension is:

$$Surface\ \text{tension=}\frac{Surface\ \text{energy}}{Area}$$
$$S.T = \frac{5\pi \times S.T}{4\pi r^2} \qquad (\text{r=radius})$$
$$\text{r}^2 = \frac{5}{4}$$
$$\text{r=}\frac{\sqrt{5}}{2}$$

Therefore, diameter is

$D = 2r$

$= 2 \times \frac{\sqrt{5}}{2}$

$= \sqrt{5} cm$

$= 2.236 cm$

Hence, the diameter is $2.236 cm$.

Question 2: Select and write the most appropriate answer from the given alternatives for each sub-question. [7]

(i) Particles in U.C.M with tangential velocity 'v' along a horizontal circle of diameter 'D'. Total angular displacement of the particle in time 't' is…..

(a) vt

(b) $\left(\frac{v}{D}\right) - t$

(c) $\frac{vt}{2D}$

(d) $\frac{2vt}{D}$

(ii) Two springs of force constants k_1 and k_2 $(k_1 > k_2)$ are stretched by same force. If w_1 and w_2 be the work done in stretching the spring then………

(a) $W_1 = W_2$

(b) $W_1 < W_2$

(c) $W_1 > W_2$

(d) $W_1 = W_2 = 0$

(iii) A and B are two steel wire and the radius of A is twice the radius of B. If they are stretched by the same load, then the stress on B is……

(a) Four times that of A
(b) Two times that of A
(c) Three times that of A
(d) Same as that of A

(iv) If sound waves are reflected from the surface of denser medium, there is phase change of……..

(a) 0 rad
(b) $\frac{\pi}{4} rad$
(c) $\frac{\pi}{2} rad$
(d) πrad

(v) A sonometer wires vibrates with frequency n_1 in air under suitable load of specific gravity $'\sigma'$. When the load is immersed in water, the frequency of vibration of wire n_2 will be….

(a) $n_1\sqrt{\frac{\sigma+1}{\sigma}}$
(b) $n_1\sqrt{\frac{\sigma-1}{\sigma}}$
(c) $n_1\sqrt{\frac{\sigma}{\sigma+1}}$
(d) $n_1\sqrt{\frac{\sigma}{\sigma-1}}$

(vi) For polyatomic molecules having 'f' vibrational modes, the ratio of two specific heats, $\frac{C_P}{C_V}$ is ……

(a) $\frac{1+f}{2+f}$
(b) $\frac{2+f}{3+f}$
(c) $\frac{4+f}{2+f}$
(d) $\frac{5+f}{4+f}$

(vii) A body of moment of inertia $5kgm^2$ rotating with the an angular velocity $6rad/s$ has the same kinetic energy as a mass of 20kg moving with a velocity of …..

(a) 5m/s
(b) 4m/s
(c) 3m/s
(d) 2m/s

Solution 2:

(i) (d)

$\frac{2vt}{D}$

(ii) (b)

$W_1 < W_2$

(iii) (a)
Four times that of A

(iv) (d)

πrad

(v) (b)

$n_1\sqrt{\frac{\sigma - 1}{\sigma}}$

(vi) (c)

$\frac{4+f}{2+f}$

(vi) (c) 3m/s

Question 3: Define linear S.H.M Show that S.H.M is a projection of U.C.M on any diameter. A metal sphere cools at the rate of . 4^0C When its temperature is 50^0C . Find its rate of cooling at the rate of 45^0C if the temperature of surrounding is 25^0C .

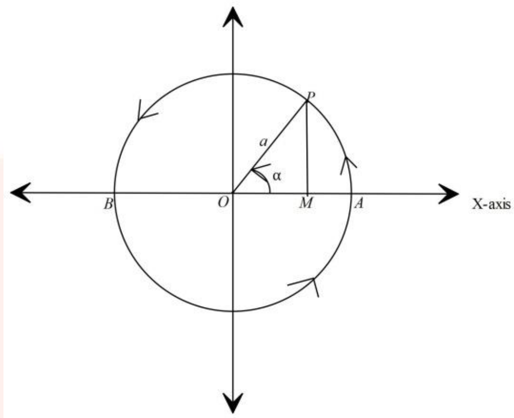
OR

Explain analytically how the stationary how the stationary waves are formed. Hence, show that the distance between node and antinode is $\frac{\lambda}{4}$.

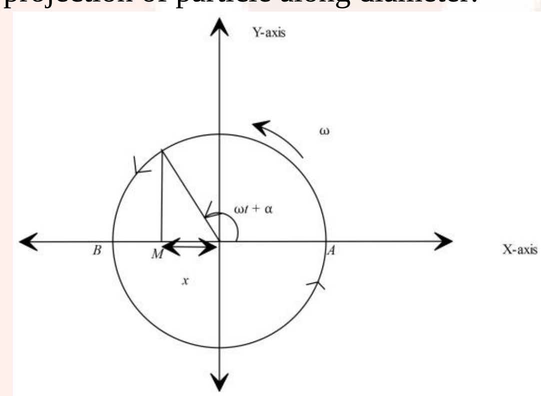
A set of 48 tuning fork is arranged in a series of descending frequency such that each fork gives 4 beats per second with preceding one. The frequency of first fork is 1.5 times the frequency of the last fork, find the frequency of the first and 42nd tuning fork.

Solution 3:

Linear S.H.M:- Linear S.H.M is the linear periodic motion of a body, in which the restoring force is always directed towards the mean position and its magnitude is directly proportional to displacement from the mean position.



(a) For an object performing U.C.M , the projection of its motion along any diameter of its path executes S.H.M.
(b) Let us consider a particle P moving along the circumference of a circle with radius 'a' and constant angular velocity 'ω' in anticlockwise
(c) 'M' is the projection of particle along diameter.



(d) Particle'P' is called the refrence particle and the circle is known as refrence circle.
(e) Since, the particle revolves, its projection along diameter AB moves back and forth. The X-component of displacement, velocity and acceleration of P is always the same as the X-component of displacement.
(f) Let us consider that the particle P starts from initial position with initial phase α
(g) In time 't' the angle Op and X-axis is $(\omega t+\alpha)$

Therefore,

$$\cos(\omega t+\alpha)=\frac{x}{a}$$

$$\therefore x=a\cos(\omega t+\alpha)......(1)$$

This is the expression of displacement at time t.

$\therefore Velocity = \frac{dx}{dy}$

$\Rightarrow V = -a\omega \sin(\omega t + \alpha)...(2)$

Therefore, acceleration $\frac{dv}{dt} = -a\omega^2 \cos(\omega t + \alpha)...(3)$

From eq 1 and 3 we get,

Acceleration $= -\omega^2 x$

Therefore, the acceleration is directly proportional to the displacement and its direction is opposite.

And, S.H.M is a projection of U.C.M along the diameter.

Now,

According to the Newton's law of cooling rate of cooling

$\frac{d\theta}{dt} = k(\theta - \theta_0)$

$k = cons\tan t$

$\theta_0 = temprature$ of surronding

Since, we have

$\frac{d\theta_1}{dt} = 4^0 C / \min$

$\theta_1 = 50^0 C, \theta_2 = 45^0 C$

$\frac{d\theta_1}{dt} = k(\theta_1 - \theta_0)$

$$\frac{\frac{d\theta_2}{dt}}{4} = \frac{45-25}{50-25}$$

$\frac{d\theta_2}{dt} = \frac{16}{5}$

$\therefore \frac{d\theta_2}{dt} = 3.2C / \min$

Hence, the rate of the cooling is $3.2C/\min$.

OR

Amplitudes of antinodes is maximum,

$A = \pm 2a$

$A = 2a\cos\dfrac{2\pi x}{\lambda}$

$\therefore \pm 2a = 2a\cos\dfrac{2\pi x}{\lambda}$

$\Rightarrow \cos\dfrac{2\pi x}{\lambda} = \pm 1$

$\therefore \dfrac{2\pi x}{\lambda} = 0, \pi, 2\pi....$

$\dfrac{2\pi x}{\lambda} = P\pi$

$\therefore x = \dfrac{P\lambda}{2} = P\left(\dfrac{\lambda}{2}\right)$

$For, x = 0, \dfrac{\lambda}{2}, \lambda, \dfrac{3\lambda}{2}, ...antinodes$

Therefore,

Distance between two successive antidotes is $\dfrac{\lambda}{2}$.

Amplitudes of nodes is zero,

$A = 2a\cos\dfrac{2\pi x}{\lambda}$

$0 = 2a\cos\dfrac{2\pi x}{\lambda}$

$\therefore \cos\dfrac{2\pi x}{\lambda} = 0$

$\therefore \dfrac{2\pi x}{\lambda} = \dfrac{\pi}{2}, \dfrac{3\pi}{2}, \dfrac{5\pi}{2}.....$

$x = (2p-1)\dfrac{\lambda}{4}$

So, the distance between any two successive modes is $\dfrac{\lambda}{4}$

We have, $n_1 = 1.5n_{48}$ and beat frequency is 5Hz.

$\therefore n_2 = n_1 - 4$
$n_3 = n_2 - 4 = n_1 - 2\times 4$
$n_{48} = n_{47} - 4 = n_1 - 47\times 4$
$n_{48} = n_1 - 188$
$n_{48} = 1.5n_{48} - 188$
$0.5n_{48} = 376$
$n_1 = 1.5n_{48} = 1.5\times 376 = 576$
$\Rightarrow n_{42} = n_{41} - 4 = n_1 - 4\times 41$
$\Rightarrow n_{42} = 400Hz$

Question 4: Attempt any three **[9]**

(i) What is the decrease in weight of a body of mass 600kg when it is taken in a mine of depth 500m?

$\left[Radius \text{ of earth is 6400km, g=9.8m/s}^2 \right]$

(ii) State and prove theorem of parallel axis about moment of inertia.

(iii) Derive Laplace's law for spherical membrane of bubble due to surface tension.

(iv) A steel wire having cross sectional area $1.5mm^2$ **when stretched by a load produces a lateral strain** $1.5\times 10^{-5.}$ **Calculate the mass attached to the wire.**

$\left[Y_{steel} = 2\times 10^{11} N/m^2, Poisson's\ ratio, \sigma = 0.291, g = 9.5m/s^2 \right]$

Solution 4:

(i) We have,

$m = 600kg$
$d = 5000m$
$R = 6400km = 6.4\times 10^6 m$
$g = 9.8m/s^2$

Weight at the surface $= mg$

Of the Earth (w_1)

$w_1 = 600\times 9.8$
$= 5880N$

Acceleration due to gravity at a depth'd' is (gd)

$gd = g\left(1-\frac{d}{R}\right)$

$gd = 9.8\left(1-\frac{5000}{6400\times1000}\right)$

$= 9.8\left(\frac{6400-5}{6400}\right)$

$= 9.8\left(1-\frac{5}{6400}\right)$

$= 9.8\times0.9992$

$\therefore gd = 9.79216$

$\therefore W_2 = mgd$

$= 600\times9.7216$

$= 5875.296N$

Therefore, Decrease in weight

$w_1 - w_2$

$= 5880 - 5875.296$

$= 4.704N$

Decrease in weight at 500m depth is 4.704N.

(ii) **Theorem of parallel axis: -** It states that the moment of inertia of a body about any axis equal to the sum of the moment of inertia of parallel axis through its center of mass and the product of their masses and the square of its perpendicular distance between them.

Proof: - Let us consider a rigid body of mass 'M' rotating about an axis passing through at 'o'.

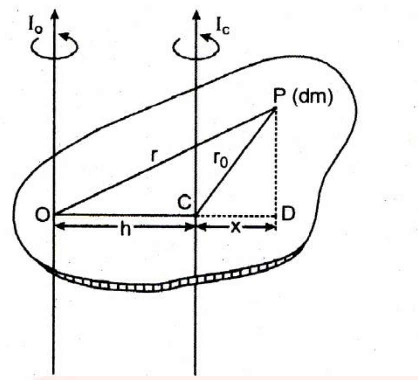
Suppose I_0 is the moment of inertia of rigid body about an axis passing through o.

Also, consider another axis parallel through first axis passing through point c and perpendicular to the plane.

I_c is the moment of inertia of the body about the second axis.

Let $oc = b$ be the distance between two parallel axes

Consider an element having mass 'dm' situated at the point p.



Therefore,

$$I_0 = \int op^2 dm$$

$$= \int r^2 dm \quad ......(1)$$

$$\therefore I_c = \int Cp^2 dm$$

$$= \int {r_0}^2 dm \quad (2)$$

Now, draw PD perpendicular to OC.

So,

$$OP^2 = OD^2 + PD^2$$

$$= (h + CD)^2 + PD^2$$

$$= h^2 + 2hCD + CD^2 + PD^2$$

$$= h^2 + 2hCD + CP^2$$

$$\therefore OP^2 = CP^2 + h^2 + 2hCD$$

Now, multiply by 'dm' in both sides and integrating both sides

$$\int r^2 dm = \int {r_0}^2 dm + \int h^2 dm + \int 2hxdm$$

$$\Rightarrow \int r^2 dm = \int {r_0}^2 dm + h^2 \int dm + 2h \int xdm \quad ....(3)$$

Since,

$$\int xdm = 0$$

As the algebraic sum of moments of all particles.

Therefore, for a body in equilibrium

$\int dm = M$

From eq$(1),(2)$ and (3)

$I_0 = I_c + Mn^2$

(iii) Let us consider a liquid drop and let the outside the pressure P_0 and inside pressure be p_i ,such that the excess pressure is $P_i - P_0$.

Now,

Let the radius of the drop increases from r to $r + \Delta r$, where Δr is very small, so that pressure inside the drop remains almost constant.

Initial surface area $(A_1) = 4\pi r^2$

Final surface area $(A_2) = 4\pi\left(r + \Delta r\right)^2$

$= 4\pi\left(r^2 + 2r\Delta r + \Delta r^2\right)$

$= = 4\pi r^2 + 8\pi r\Delta r + 4\pi\Delta r^2$

As Δr is very small, Δr^2 is neglected $4\pi\Delta r^2 \approx 0$

Increase in surface area $(dA) = A_2 - A_1 = 4\pi r^2 + 8\pi r\Delta r - 4\pi r^2$

So, increase in surface area $(dA) = 8\pi r\Delta r$

Work done to increase the surface area $(dA) = 8\pi r\Delta r$ is extra energy.

So,

$dW = TdA$

$dW = T \times 8\pi r\Delta r \quad .....(1)$

This is equal to the product of force and distance Δr .

$dF = \left(P_i - P_0\right)4\pi r^2$

So, the increase in radius of the bubble is Δr .

$dW = dF\Delta r = \left(Pi - P_0\right)4\pi r^2 \times \Delta r \quad .....(ii)$

From eq(i) and(ii), we get

$\left(Pi - P_0\right)4\pi r^2 \times \Delta r = T \times 8\pi r\Delta r$

$\left(Pi - P_0\right) = \dfrac{2T}{r}$

Hence, this is the Laplace theorem.

(iv) We have,

$A = 11.5mm^2 = 1.5\times10^{-6} m^2$

Lateral strain 1.5×10^{-5}

$Y_{steel} = 2\times10^{11} N/m^2$
$\sigma = 0.291$
$g = 9.8m/s^2$

Since,

$$\sigma = \frac{lateral \text{ strain}}{longitudinal \text{ strain}}$$

So,

Longitudinal strain $= \dfrac{1.5\times10^{-5}}{0.291} = 5.154\times10^{-5}$

$$\gamma = \frac{longitudinal \text{ stress}}{longitudinal \text{ strain}}$$
$$= \frac{force}{Area\times strain}$$
$$= \frac{Mg}{A\times5.154\times10^{-5}}$$
$$\Rightarrow M = \frac{2\times10^{11}\times1.5\times10^{-6}\times5.154\times10^{-5}}{9.8}$$
$$= \frac{2\times1.5\times5.154}{9.8}$$
$$= \frac{115.462}{9.8}$$
$$\therefore M = 1.578Kg$$

SECTION -II

Question 5: Attempt any six **[5]**

(i) What is diffraction of light? Explain its two types.

(ii) Draw a neat labeled diagram for the construction of ‘cyclotron’.

(iii) Distinguish between ‘paramagnetic’ and ‘ferromagnetic’ substances.

(iv) Write a short not on surface wave propagation of electromagnetic waves.

(v) The combined resistance of galvanometer of resistance 500Ω and its shunt is 21Ω. Calculate value of shunt.

(vi) The susceptibility of magnesium at 200k is 1.8×10^{-5}. At what temperature will the susceptibility decrease by 6×10^{-6}?

(vii) The co-efficient of mutual induction between primary and secondary coil is 2H. Calculate induced e.m.f. If current of 4A is cut off in 2.5×10^{-4} second.

(viii) The decay constant of radioactive substance is 4.33×10^{-4} per year. Calculate its half-life period.

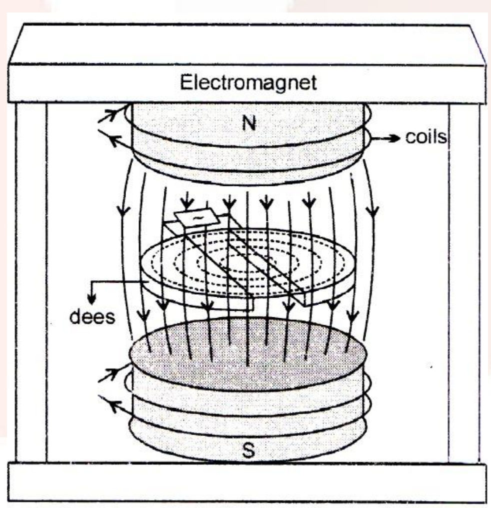
Solution 5:

(i) **Diffraction of light:-** Bending of light near the edge of an obstacle or slit and spreading into the region of geometrical shadow.

The types of diffraction of light are as follows:

(1) **Fraunhofer diffraction:-** The source of light and the screen on which the diffraction pattern is obtained are effectively at infinite distance from the diffracting system.
(2) Fresnel diffraction:- The source of light and the screen are kept at finite distance From the diffracting system.

(ii)



(iii) The difference between paramagnetic and ferromagnetic substance are as follows:

Paramagnetic substances	**Ferromagnetic substances**
(a) Those substances which are weakly attracted by a magnet are known as paramagnetic substance.	(I) Those substances which are strongly attracted by a magnet are known as paramagnetic substance.
(b) Those materials lose their magnetism on removal of the external field. And cannot used to make permanent magnet.	(II) Those materials retain some magnetism on removal of the external field. And can be used to make permanent magnet.
(c) The susceptibility is positive and very small.	(III) The susceptibility is positive and very high.
(d) Some paramagnetic substances are chromium, aluminum, manganese.	(IV) Some ferromagnetic substances are Iron, nickel, cobalt, gadolinium, and dysprosium.

(iv) Surface wave propagation of electronic wave can be determine as the radio waves from the transmitting antenna propagate along surface of the Earth to reach the receiving antenna.

(1) Electromagnetic waves which are vertically polarized can travel in this mode.
(2) The horizontal component of electric field in contact with Earth is short circuited.
(3) These are used to local broadcasting e.g. ship communication and radio navigation. For TV and FM signals.

(v)

We have,

Combined resistance

$R = 21\Omega$
$G = 500\Omega$

So, the value of shunt:

The galvanometer and shunt are connected in parallel.

$$\frac{1}{R}=\frac{1}{G}+\frac{1}{S}$$
$$R=\frac{GS}{G+S}$$
$$21=\frac{500s}{500+S}$$
$$21\times 500+21\times S=500s$$
$$10500=500s-21s$$
$$\therefore s=\frac{10500}{479}$$
$$\therefore S=21.93\Omega$$

Hence, the value of shunt is 21.93Ω.

(vi)

We have,

$$X_1=1.8\times 10^{-5}$$
$$T_1=200k$$

Susceptibility decreases by 6×10^{-6}

$$\therefore X_2=X_1-6\times 10^{-6}$$
$$=1.8\times 10^{-5}$$
$$=0.6\times 10^{-5}$$
$$X_2=1.2\times 10^{-5} \quad ......(1)$$

Magnetic susceptibility is inversely proportional to the Absolute temperature,

$$X\propto\frac{1}{T}$$
$$X_1T_1=X_2T_2$$
$$1.8\times 10^{-5}\times 200=1.2\times 10^{-5}T_2$$
$$T_2=\frac{31.8\times 200T_2}{1.2}$$
$$T_2=300k$$

Hence, the susceptibility decreases by 6×10^{-6} $300k$.

(vii)

We have,

$M=2H$

$I_1=4A$

$I_2=0A$

$t=2.5\times10^{-4}s$

According to definition,

$$e=\frac{-d\phi}{dt}$$
$$=-M\,\frac{di}{dt}$$
$$=M\,\frac{(I_2-I_1)}{t}$$
$$=\frac{-2(0.4)}{2.5\times10^{-4}}$$
$$=\frac{2\times4\times10}{25\times10^{-4}}$$
$$=\frac{16}{5}\times10^4$$
$$e=3.2\times10^4V.$$

Hence, the induced emf is 3.2×10^4V.

(viii) We know the half-life period of a radioactive substance is given by,

$$T=\frac{0.693}{\lambda}$$
$$\therefore T=\frac{0.693}{4.33\times10^{-4}}$$
$$\therefore T=0.1601\times10^4\,years$$
$$T=1601years$$

Hence, the half-life is $1601years$.

Question 6: Select and write the most appropriate answer from the given alternative for each sub-question: [7]

(i) If the polarizing angle for a given medium is 60^0, then the refractive index of the medium is ….

(a) $\frac{1}{\sqrt{3}}$

(b) 1

(c) $\sqrt{\frac{3}{2}}$

(d) $\sqrt{3}$

(ii) The resolving power of a telephone depends upon the ……..

(a) Length of the telescope
(b) Focal length of an objective
(c) Diameter of an objective
(d) Focal length of an eyepiece

(iii) Electric intensity due to a charged sphere at a point outside the sphere decrease with……

(a) Increase in charge on sphere.
(b) Increase in dielectric constant
(c) Decrease the distance from the center of sphere.
(d) Decrease in square of distance from the center of sphere.

(iv) In potentiometer experiment, if l_1 is the balancing length for e.m.f of cell internal resistance r and l_2 is the balancing length for its terminal potential difference when shunted with resistance R then:

(a) $l_1 = l_2\left(\frac{R+r}{R}\right)$

(b) $l_1 = l_2\left(\frac{R}{R+r}\right)$

(c) $l_1 = l_2\left(\frac{R}{R-r}\right)$

(d) $l_1 = l_2\left(\frac{R-r}{R}\right)$

(v) The energy of photon of wavelength λ is …….

[h=Plank's constant, c=speed of light in vacuum]

(a) hcλ

(b) $\frac{h\lambda}{c}$

(c) $\frac{\lambda}{hc}$

(d) $\frac{hc}{\lambda}$

(vi) Which logic gate corresponds to the truth table given below?

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(a) AND
(b) NOR
(c) OR
(d) NAND

(vii) The processes of superimposing a low frequency signal on a high frequency wave is ………

(a) Detection
(b) Mixing
(c) Modulation
(d) Attenuation

Solution 6:

(i) d

$\sqrt{3}$

(ii) (c) Diameter of an objective

(iii) (b) Increase in dielectric constant

(iv) (a)

$$l_1 = l_2\left(\frac{R+r}{R}\right)$$

(v) (d)

$$\frac{hc}{\lambda}$$

(vi) (b) NOR

(vii) (c) Modulation

Question 7.

State the principle on which transformer works. Explain its working with construction. Derive an expression for ratio of e.m.f and currents in terms of number of turns in primary and secondary coil.

A conductor of any shape, having are $40cm^2$ placed in air is uniformly charged with a charge $0.\mu C$. Determine the electric intensity at a point just outside its surface. Also, find the mechanical force per unit area of the changed conductor [$\in_0 = 8.85\times10^{-12}\,S.Iunits$]

OR

With the help of a neat labeled diagram, describe the Geiger –Marsden experiment. What is mass defect?

The photoelectric work function for a metal surface is 2.3 eV. If the light of wavelength 6800 A^0 is incident on the surface of metal, find threshold frequency and incident frequency. Will there be an emission of photoelectrons or not?

[Velocity of light c=$3\times10^8 m/s$, Plank's constant, $h = 6.63\times10^{-34}$ Js]

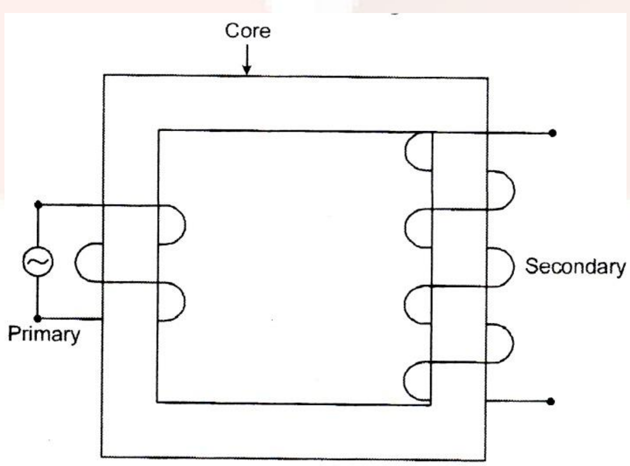
Solution 7:

The value of alternating voltage can be changed from high to law value at varius stage by using a device called transformer.

Principle:- It is based on mutual induction that is production of induced emf in one coil due to change of current in neighboring coil.

Construction:-

It consists of two coils, primary (p) and secondary (s) insulated from each other and around on a soft iron core.



Primary is the input coil and secondary coil is the output coil.

Working:

(a) When an alternating voltage is applied to the primary coil, the current through the coil goes on changing.
(b) Magnetic flux also changes.
(c) An emf is induced in each of two coils.
The amount of magnetic flux depends upon the number of turns of the coil.

Derivation: Let us suppose ϕ be the magnetic flux per turn linked with both the coils at certain instant 't'.

$N_p\phi$ is the magnetic flux linked with the primary coil at a certain instant 't'.

$N_s\phi$ is magnetic flux linked with the secondary coil at certain instant 't'.

Where, N_p N_s

$$\therefore e_q = \frac{-d\phi_p}{dt}$$

$$= -N_p \frac{-d\phi_p}{dt} \quad ....(1)$$

$$e_s = \frac{-d\phi_s}{dt}$$

$$= -N_s \frac{-d\phi_s}{dt} \quad ......(2)$$

By taking ratio of (1) and (2)

$$\frac{e_s}{e_p} = \frac{N_s}{N_p} \quad .....(3)$$

It is known as the equation of the transformer.

Therefore, For an ideal transformer,

Input power is equal to output power,

$$e_p i_p = e_s i_s$$

$$\frac{e_s}{e_p} = \frac{i_p}{i_s} \quad ......(4)$$

$\therefore$ from eq(3) and (4)

$$\therefore \frac{e_s}{e_p} = \frac{N_s}{N_p}$$

This is the ratio of emf.

Numerical:

We know that the force per unit area of the conductor is

$$\sigma = \frac{v}{A}$$
$$\sigma = \frac{2\times10^{-7}}{40\times10^{-4}}$$
$$= \frac{1}{2}\times\frac{10^{-7}}{10^{-4}}$$
$$\Rightarrow \sigma = 0.5\times10^{-4}$$

Now, by the Gauss theorem, the electricity outside the charged conductor:

$$E = \frac{\in}{\in_0}$$
$$= \frac{5\times10^{-5}}{8.85\times10^{-12}}$$
$$= \frac{10^{7}}{1.77\times1}$$
$$= 0.5650\times10^{7}$$
$$= 5.65\times10^{6}\, N / mC$$

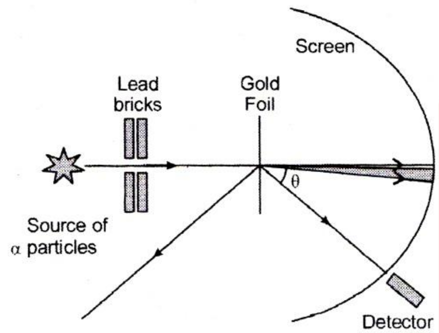
And the mechanical force per unit area of the charged conductor

$$F = \frac{\sigma^2}{2\in_0}$$
$$\therefore F = \frac{\left(5\times10^{-5}\right)^2}{2\times8.85\times10^{-12}}$$
$$= \frac{25\times10^{-10}}{17.7\times10^{-12}}$$
$$= 141.2 N / m^2$$

Hence, the electricity outside the conductor is $5.65\times10^{6}\, N / mC$ and the mechanical force is $141.2 N / m^2$.

OR

Geiger –Marsden experiment: IN this experiment a narrow beam of $\alpha-$particles from radioactive sources was incident on a gold foil. And the wave is detected by the detector fixed on the rotating stand. The detector used zinc sulphide screen.



They observed the number of $\alpha-$particles as a function of scattering angle. The angle deviation (θ) of $\alpha-$particles from its original direction. Then they show that most of the $\alpha-$particles passed undeviated and only few scattered by more than 1 degree. And some are deflected a lightly and only few were deflected by 90degree, and some were bounce back with 180degrees.

Mass defect:

It is observed that the mass of a nucleus is smaller than the sum of the masses of the constituent nucleons in the Free State. The difference between the actual mass of the nucleus and the sum of the masses of constituent nucleons is called mass defect.

The mass defect is

$$\Delta m=\left[Zm_p+(A-Z)m_p\right]-M$$

Where, Z is the atomic number (number of protons), A is the mass number, (A-Z) is the number of neutrons, m_p is the mass of a proton, m_n is the mass of a neutron and M is the measured mass of a nucleus.

Problems:

Given:

$$\phi_0=2.3eV=2.3\times1.6\times10^{-19}J=3.68\times10^{-19}J$$

$$\lambda=6800,\overset{o}{A}=6800\times10^{-10}m,c=3\times10^{8}m/s$$

$$h=6.63\times10^{-34}Js$$

We know that the incident frequency is given as

$$v = \frac{c}{\lambda}$$

Thus,

$$v = \frac{3\times10^{8}}{6800\times10^{-10}} = 4.41\times10^{14}\,\text{Hz}$$

Now, if the incident frequency is greater than the threshold frequency, then photoelectrons will be emitted from the metal surface. The threshold frequency is given from work function as

$$v_0 = \frac{\phi_0}{h}$$

$$\Rightarrow v_0 = \frac{3.68\times10^{-19}}{6.63\times10^{-34}} = 5.55\times10^{14}\,\text{Hz}$$

Qusetion.8 Attempt any three: [9]

(i) Determine the change in wavelength of light during its passage air to glass. If air to glass. If the refractive index of glass with respect to air is 1.5 and the frequency of light is $3.5\times10^{14}\,Hz$, **find the wave number of light in glass.[Velocity of light in air (c=**$3\times10^{8}\,m/s$, **Planck's constant,** $h = 6.63\times10^{-34}\,Js$ **].**

(ii) In biprism experiment, 10th dark brand is observed at 2.09 mm from the central bright point on the screen with red light of wavelength 6400 A^0 **. By how much will fringe width change if blue light of wavelength 4800** A^0 **is used with the same setting?**

(iii) Describe kelvin's method to determine the resistance of galvanometer by using Meter Bridge.

(iv) Explain the elementary idea of an oscillator with the help of block diagram.

Solution 8:

(i) We have,

Frequency, $(n) = 3.5\times10^{14}\,Hz$

RI of the glass $= 1.5$

$c = 3\times10^{8}\,m/s$

So, the wave number of the glass is:

$c = \frac{\lambda}{n}$

$\lambda = \frac{c}{n}$

Therefore, the wavelength

$$\lambda_a = \frac{3\times10^8}{3.5\times10^{14}}$$
$$= \frac{6}{7}\times10^{-6}$$
$$= 0.857\times10^{-6} m$$
$$8571\dot{A}$$

Now, by definition,

$$\eta = \frac{\lambda_a}{\lambda_g}$$
$$\lambda_g = \frac{\lambda_a}{\eta}$$
$$= \frac{8571}{1.5}$$
$$= 5715\dot{A}$$

So, the change in the wavelength of light is

$$\lambda_a - \lambda_g$$
$$= 8571 - 5715$$
$$= 2856\dot{A}$$

Therefore, the wave number in the glass is $= \frac{1}{\lambda_g}$

$$\bar{\lambda} = \frac{1}{5.715\times20^{-10}}$$
$$= 0.175\times10^7$$
$$\therefore \bar{\lambda} = 1.75\times10^6 \; per.m$$

Hence, the changer in wavelength is $2856\dot{A}$ and the wave number is $1.75\times10^6 \; per.m$.

(ii) We have,

10^{th} dark band is observed at 2.09mm from the central point.

$\lambda_r = 6400\dot{A}, \lambda_b = 4800\dot{A}$

So, the position of 10^{th} dark band is observed at 2.09mm from the central point is given by:

$$X_{10} = \frac{\left(2\times10^{-1}\right)}{2}\frac{\lambda D}{d}$$

$$2.09\times10^{-3} = \frac{19}{2}\frac{\lambda D}{d}$$

$$\therefore \frac{\lambda D}{d} = \frac{2.09\times10^{-3}\times2}{19}$$

$$\frac{\lambda_r D}{d} = \frac{4.18}{19}0^{-3}$$

$$= 0.22\times10^{-3}m$$

It is the position.

Since, bandwidth is directly proportional to the wavelength

$$X \propto \lambda$$

$$\therefore = \frac{X_r}{X_b} = \frac{\lambda_r}{\lambda_b}$$

$$\frac{0.22\times10^{-3}}{X_b} = \frac{6400}{4800}$$

$$\Rightarrow X_b = \frac{0.22\times10^{-3}\times4800}{6400}$$

$$= 0.165\times10^{-3}m$$

Therefore, the change is

$$X_r - X_b$$

$$= 0.22 - 0.165$$

$$= 0.055mm$$

Hence, the fringe width changes by $0.055mm$.

(iii) G is the galvanometer, R is the resistance from the resistance box, Ac is the metal wire of one meter, R_n is the rheostat, E is cell, K is the plug key and K' is jockey.

We have to take these apparatus,

(1) The galvanometer has resistance is to be determined and connected in in gap and a resistance box in another gap of meter-bridge.
(2) A cell key and rheostat which are connected in series with wire.
(3) The junction of galvanometer and resistance box which is connected to jockey.

Working,

(I) We have to close the key and take the suitable resistance R from the resistance box.

(II) The jockey is touched to different points on the wire A point 'D' for which galvanometer shows same deflection.

(III) The point 'D' is known as the balance point

(IV) Let us consider $l(AD) = l_g$
$l(CD) = l_r$

Since, the bride is balanced,

$$\frac{G}{R} = \frac{REsisi\tan ce \text{ of } l_g}{\mathrm{Re}\,sis\tan ce \text{ of } l_r}$$

Let σ is the resistance per unit length of wire AC.

$$\therefore \frac{G}{R} = \frac{\sigma l_g}{\sigma l_r}$$

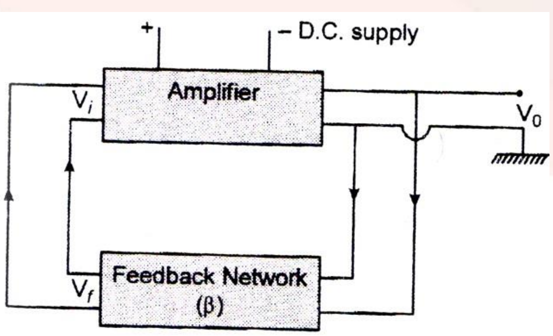
$$\Rightarrow \frac{G}{R} = \frac{l_g}{l_r}$$

$$\Rightarrow G = R \cdot \frac{l_r}{l_r}$$

$$G = \frac{Rl_g}{100 - l_g}$$

Hence, the unknown resistance of galvanometer can be determined.

(iv) An oscillator is the device which generates oscillation of desired frequencies. It is used in radio/TV receivers sets.



Oscillator requires an amplifier and a feedback network with frequency determine by components. When a part of the amplifier is coupled to the input of the amplifier is known as feedback. If the feedback is out of phase with the input, it is known as magnetic feedback.

$$A_f = \frac{A}{1 - A\beta}$$

Where, A_f is the voltage gain with feedback, A is the voltage gain without feedback β is the feedback factor.

If the some frequency $A\beta = 1$, then the system gain becomes ∞ and the circuit begins oscillate at that frequency. $A\beta = 1$ is known as Barkhausen criterion.

The frequency of oscillation depends on the LC or RC combination used in feedback network. When the supplied power to the circuit is turned ON then the wide range of frequency is generated in the circuit.